



الكلية الجامعية للعلوم التطبيقية
University College of Applied Sciences

Project title:

Image Blur vs. Sharp Classification Using Traditional Neural Networks

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Introduction

Detecting blur in images is an important problem in computer vision. It is especially for systems that assess and enhance image quality. Images may become blurred because of motion or focus issues. Which reduces edge clarity and fine details.

In this project, we design a neural network to classify images into sharp and blurred categories. Instead of using raw pixel values, we extract meaningful features such as edge strength, gradient size, contrast and high-frequency energy, from. These features are used to train a Multi-Layer Perceptron (MLP) classifier.

The objective of this project is to evaluate how effectively neural networks combine with handcrafted features can classify blurred images.

Theoretical Background

Artificial neural network (ANNs) are computational models inspired by the way the human brain processes information. They are composed of layers of connected neurons, where each neuron receives inputs, applies weights, and produces an output. Neural networks are widely used in classification tasks because they can learn patterns from data through training.

In this project, we use a Multi-Layer Perceptron (MLP), which is a type of feedforward neural network composed of an input layer, one or more hidden layers, and an output layer. Each neuron applies a weighted sum of its inputs followed by an activation function.

The activation function introduces non-linearity into the network, allowing it to learn complex decision boundaries. In our model, we use the ReLU activation function in the hidden layers.

To train the network, we minimize a loss function that measures the difference between predicted and true labels. For classification tasks, the model optimizes the cross-entropy loss.

The optimization process is performed using gradient-based methods such as Adam and Stochastic Gradient Descent (SGD), which update the network weights iteratively to reduce the loss.

Methodology

• Dataset

The dataset used in this project was obtained from Kaggle. It contains 1050 images (350 triplets), where each triple represents the same scene captured in three different conditions: sharp, defocused-blurred, and motion-blurred images.

For the classification task, motion blur and defocus blur images were combined into a single Blur class, while sharp images were kept as a separate class.

To keep the training process efficient, a subset of the dataset was used. Specifically, 300 sharp images and 600 blurred images (300 motion and 300 defocus blur) were selected for training and evaluation.

Source: Kaggle Blur Dataset

<https://www.kaggle.com/datasets/kwentar/blur-dataset>

• Preprocessing

All images were converted to grayscale to reduce computational complexity and focus on structural features rather than color information. Each image was resized to a fixed size of 256 * 256 pixels to ensure consistency across the dataset.

• Feature Extraction

Instead of using raw pixel values, handcrafted features extracted from each image to represent its structural and frequency characteristic. The extracted features include:

- Variance of Laplacian: measures edge sharpness.
- Tenengrad (Sobel gradient magnitude): captures edge strength.
- Image Contrast: measures intensity variation.
- High-Frequency Energy (FFT): captures frequency-domain sharpness information.

These features form a numerical representation of each image and are used as input to the neural network.

- **Model Architecture**

A Multi-Layer Perceptron (MLP) classifier was used for the classification task. The network consists of:

- Input layer (4 features).
- Two hidden layers with ReLU activation.
- Output layer with two neurons (Sharp / Blur).

The model was trained using different optimization algorithms to evaluate performance differences.

- **Training Setup**

The dataset was split into training and testing sets. Feature normalization was applied using a standard scaler before training to improve convergence.

The model was trained for a fixed number of iterations, and performance was evaluated using accuracy as the primary metric. Additionally, precision, Recall, F1-score, and confusion matrix were analyzed to better understand classification behavior between sharp and blur classes.

Experiments and Results

To evaluate the proposed model, several experiments were conducted using different optimization algorithms. The dataset was divided into training and testing sets, and feature normalization was applied before training.

The primary evaluation metric was accuracy. In addition, precision, Recall, F1-score, and confusion matrix.

Adam Optimizer

When using the Adam optimizer, the model achieved an accuracy of approximately 94% on the test set. The confusion matrix showed that most blurred images were correctly classified, while a small number of sharp images were misclassified as blur.

The results indicate that Adam provided stable convergence and strong classification performance.

SGD Optimizer

When using SGD, the model achieved a lower accuracy of approximately 91%. Although the performance was still acceptable, the model showed slightly more misclassification in the sharp class compared to Adam.

This suggests that Adam optimizer was more effective for this problem.

Discussion

The experiment results show that the extracted features combined with an MLP model can effectively classify sharp and blurred images. Most errors occurred in sharp images with weaker edge information. The slight dataset imbalance may also have influenced the predictions. Overall, the Adam optimizer achieved better performance than SGD.

Conclusion

In this project, a neural network model was developed to classify images as sharp or blurred using handcrafted spatial and frequency features. The MLP classifier achieved high accuracy, with Adam providing the best results. These findings demonstrate that feature engineering combined with neural networks is an effective approach for blur detection.